

Ziba Hatamian

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Machine Learning Engineer specializing in end-to-end ML systems—from experiment design and model optimization to API development and Dockerized deployment. Experienced in building reproducible ML pipelines using PyTorch, FastAPI, Docker, and MLflow, with a strong emphasis on systematic experimentation and error analysis.

TECHNICAL SKILLS

Python, SQL | PyTorch, Scikit-Learn, CNN, FFN | CV: TorchVision, Augmentation | NLP: TF-IDF, Text Classification

MLOps: MLflow, FastAPI, Docker, Git, REST API | Experiment Mgmt, Threshold Optimization, Error Analysis

PROJECTS

Heart Disease Prediction — End-to-End MLOps Pipeline | PyTorch, Sklearn, MLflow, FastAPI, Docker

- Complete pipeline: preprocessing → deployment | Compared Logistic Regression vs Neural Network
- Threshold optimization using validation F1-score | Used Skorch for PyTorch+Sklearn integration
- *Achieved **81% accuracy**, **0.89 ROC-AUC**, and **0.81 F1-score**, with **93% recall** for heart disease detection* | MLflow tracking + Dockerized FastAPI

EMNIST Handwritten Character Recognition | PyTorch, CNN, Experiment Management, Error Analysis

- Experiment-driven framework with configurable architectures, optimizers, schedulers
- 10+ systematic experiments: batch norm, dropout, kernel size, Adam/AdamW, schedulers
- **95.30% test accuracy (↑ from 93.46%)** | Custom error analysis: 'l' vs 'i', 'g' vs 'q' misclassifications

Twitter Airline Sentiment Analysis | NLP, Scikit-Learn, TF-IDF

- Multi-class classifier: Naive Bayes, Logistic Regression, LinearSVC | **0.77 accuracy, 0.70 Macro F1**

Advertising Sales Prediction API | Regression, FastAPI, Docker, MLflow

- Built a production-ready regression API using FastAPI, Docker and MLflow for experiment tracking and real-time inference.

EDUCATION

M.Sc. Biotechnology — 2017 | B.Sc. Genetics | Transitioned to Data Science (2023–2026)

DESIGN PHILOSOPHY: Modularity | Systematic Experimentation | Error Analysis | Reproducibility

GITHUB: github.com/zziibbaa/ML_Portfolio — 13 End-to-End Projects: CV • NLP • MLOps